

質因數分解(佔分 20 分)

問題描述：

質因數是能整除給定正整數的質數，試撰寫一個程式進行質因數分解，就是把一個數字，切分為數個質數的乘積。顯示質因數分解結果時，輸出質因數由小至大排列，每個質因數以空格為區隔。例如 48 的質因數分解 $2*2*2*2*3$ ，表示成 2 2 2 2 3。

輸入說明：

第 1 行輸入 k 筆欲進行質因數分解的正整數，其中 $1 \leq k \leq 25$ 。

第 2 行至 1+k 行輸入正整數 m，其中 $2 \leq m \leq 1000000$ 。

輸出說明：

根據每行輸入正整數 m，輸出其質因數分解結果。

範例 1：

輸入：

2

16

17

輸出：

2 2 2 2

17

範例 2：

輸入：

2

36

999997

輸出：

2 2 3 3

757 1321

Prime Factorization (20 points)

Problem Description:

A prime factor is a prime number that can perfectly divide a given positive integer. Write a program to perform prime factorization, which involves breaking a number down into a product of its prime factors. When displaying the results, the prime factors should be listed in ascending order, separated by spaces. For example, the prime factorization of 48 is $2*2*2*2*3$, which is represented as 2 2 2 2 3.

Input Description:

Line 1: An integer k ($1 \leq k \leq 25$), representing the number of positive integers to be factorized.

Lines 2 to $1+k$: A positive integer m ($2 \leq m \leq 1000000$).

Output Description:

For each input integer m , output its prime factorization results.

Example 1:

Input:

2
16
17

Output:

2 2 2 2
17

Example 2:

Input:

2
36
999997

Output:

2 2 3 3
757 1321